Stroke

CLINICAL AND POPULATION SCIENCES

Late-Life Incident Stroke in the Atherosclerosis Risk in Communities Study: Etiology and Prediction

Jing Wang, MHS; Marco Egle, PhD; Zhenghao Jin, MHS; Kamakshi Lakshminarayan, MD, PhD; Chiadi E. Ndumele, MD, PhD; Josef Coresh, MD, PhD; Rebecca F. Gottesman, MD, PhD; Michelle C. Johansen, MD, PhD

BACKGROUND: As life expectancy rises, identifying causes and risk factors for incident acute ischemic stroke (AIS) among the oldest-old (≥80 years) is increasingly important. We examined whether the effect of age at stroke on AIS subtype is mediated by embolic risk factors and whether these factors improve AIS prediction.

METHODS: Stroke-free participants from the ARIC study (Atherosclerosis Risk in Communities) who developed AIS between visit 5 (2011–2013) and visit 10 (2023) were included for causal analysis; Stroke-free participants at visit 5 were included for prediction analysis. In logistics regression models, the association between age at stroke-onset (≥80 versus <80 years) and adjudicated AIS subtype (embolic ischemic stroke versus thrombotic ischemic stroke) was determined. Bootstrapped mediation analyses (1000-iterations) tested whether atrial fibrillation, myocardial infarction, coronary heart disease, heart failure, and electro/echocardiogram measures mediated the age-AIS subtype relationship. C statistics were calculated for AIS prediction (Predicting Risk of Cardiovascular Disease Events, CHA_2DS_2-VASc) and compared preinclusion and postinclusion of embolic risk factors.

RESULTS: Of 6213 stroke-free participants at visit 5, 277 (4.4%) developed AIS during a median (Q1–Q3) of 5.1 (2.6–7.1) years (median [Q1–Q3] age: 76 [72–80] years; median [Q1–Q3] age at AIS: 81 [77–86] years; 62% female; 99 embolic ischemic stroke and 178 thrombotic ischemic stroke). Individuals with AIS ≥80 years had higher odds of embolic ischemic stroke (versus thrombotic ischemic stroke) compared with those aged <80 years (odds ratio, 1.90 [95% CI, 1.09–3.31]). The effect of age at stroke-onset on embolic ischemic stroke was mediated by atrial fibrillation (44%; P=0.03), an abnormal left atrium volume index (45%; P=0.048), or an abnormal P-wave axis (43%; P=0.04). The predictive performance for AIS ≥80 years using the Predicting Risk of Cardiovascular Disease Events equation (N=5702, C statistic, 0.49 [95% CI, 0.45–0.53]), or CHA_2DS_2-VASc score (N=5739, C statistic, 0.57 [95% CI, 0.55–0.59]) was poor, but inclusion of embolic risk factors improved the performance (Predicting Risk of Cardiovascular Disease Events: C statistics, 0.77 [95% CI, 0.74–0.80]; CHA_2DS_2-VASc: C statistics, 0.63 [95% CI, 0.59–0.67]).

CONCLUSIONS: These findings suggest that identification and control of embolic risk factors are critical to reduce stroke risk as people age, and better stroke-specific prediction tools are needed.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: atherosclerosis ■ atrial fibrillation ■ ischemic stroke ■ myocardial infarction ■ risk factors

Approximately 75% of all strokes occur in individuals aged ≥65 years, and the risk of incident stroke doubles every decade after age 55 years.[1] As life expectancy increases, stroke risk is anticipated to rise, but the reason an individual might not experience an incident acute ischemic stroke (AIS) until age ≥80 years (oldest-old) is not well understood.

It is likely that stroke risk factors differ between those who have their AIS at a younger age versus those who have their AIS at an older age. Traditional risk factors, such as hypertension, diabetes, and obesity, that are predictive of AIS in middle-age or older age, may carry a different risk of stroke among the oldest-old.[2,3] Cardiac conditions such as atrial fibrillation (AF) and left

Correspondence to: Michelle C. Johansen MD, PhD, Department of Neurology, The Johns Hopkins University School of Medicine, Phipps 4th floor, Ste 477, 600 N Wolfe St, Baltimore, MD 21287. Email mjohans3@jhmi.edu

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Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
AIS	acute ischemic stroke
ARIC	Atherosclerosis Risk in Communities
EIS	embolic ischemic stroke
LDL	low-density lipoprotein
PREVENT	Predicting Risk of Cardiovascular Disease Events
TIS	thrombotic ischemic stroke

ventricular hypertrophy were found to be significantly associated with stroke risk across age but appear more important for AIS in the elderly.[2] The importance of both clinical and preclinical cardiac disease may be more prominent as stroke mechanisms as individuals age, as they can emerge later in life. It is unclear which embolic risk factors matter the most in causing AIS among the oldest-old.

Existing stroke risk prediction tools, such as the Predicting Risk of Cardiovascular Disease Events (PREVENT) equation, have focused on leveraging traditional stroke risk factors that are known to be important in midlife.[4] Given the potential importance of embolic risk factors in those who survive to have a first stroke in their 80s, adding embolic risk factors may better predict stroke in this age group.

Therefore, we aim to leverage a community-based cohort of older adults with ongoing vascular risk factor data and stroke surveillance, to determine if there is a higher risk of having an embolic ischemic stroke (EIS), among those individuals who experience their incident AIS at age ≥80 years as compared with those who experience their incident AIS at age <80 years. If an association is found, we aim to assess whether the effect of age at AIS on stroke subtype is mediated through embolic risk factors. Finally, we aim to assess if AIS prediction improves by adding embolic risk factors to current stroke prediction tools.

METHODS

Written informed consent was obtained from all ARIC study (Atherosclerosis Risk in Communities) participants included in the study, and the appropriate institutional review boards approved the study. The study followed the STROBE reporting guidelines (Strengthening the Reporting of Observational Studies in Epidemiology; Supplemental Material).[5] The data used in this study can be made available per ARIC study guidelines.

Study Population

The ARIC study is a population-based prospective cohort study of 15 792 participants aged 45 to 64 years recruited from 4 US communities (Forsyth County, NC; suburbs of Minneapolis, MN; Washington County, MD; Jackson, MS) during 1987 to 1989, and who have since attended up to 11 study visits, with ongoing study visits and active surveillance of events including stroke.[6,7] The overall response rate for follow-up among ARIC participants has been excellent at over 90%, with 54% of Black participants from the original cohort and 62% of White participants remaining in the study by visit 5 (2011–2013), considering attrition due to death.[8] More detailed methods regarding the initial ARIC participant enrollment have been previously described.[7] Briefly, information regarding vascular and embolic risk factors was collected at study baseline, each subsequent study visit, annual phone calls until 2011, and then semiannual phone calls.[7]

This study uses visit 5 (age range, 67–90 years) as the baseline, given that the primary aim of the study is to characterize AIS occurring at age ≥80 years. Participants were followed until loss to follow-up or end of visit 10 (2023, age range, 78–99 years). Participants who were neither Black nor White, and those of the Black race in Maryland or Minnesota, were excluded due to small numbers. Stroke-free participants at visit 5 who underwent ECG at visit 5, or transthoracic echocardiography at visits 5 or 7 (2018–2019) were eligible for inclusion in this study. Participants who subsequently developed a definite or probable adjudicated ischemic stroke during follow-up with complete covariates were included in the causal analysis. For the prediction analysis, all stroke-free participants at visit 5 with complete covariates were included (Figure 1).

Stroke Subtype

Stroke events are actively surveilled in ARIC using study visits, annual and semiannual phone interviews, and through hospital record review. Hospital-related potential stroke is based on hospitalized discharge data, and stroke-related deaths are collected through the National Death Index.[9] Stroke events are initially classified based on a computer-based algorithm using the validated criteria from the National Survey of Stroke, then are verified and adjudicated by physicians as definite or probable (categories used for this analysis), possible, or no stroke, with any difference between the algorithm and the physician reviewer triggering a second review.[6] Detailed ARIC stroke event criteria have been described previously.[6] Definite stroke events require brain imaging confirmation (or autopsy), and probable stroke events lacked exclusionary findings on imaging but had an appropriate clinical presentation.[10] Ischemic stroke events are then categorized into EIS or thrombotic ischemic stroke (TIS), with greater details described in the Supplemental Methods.[6,11] TIS are further categorized based on stroke lesion size and neuroimaging reports evaluated by physicians into lacunar, carotid, or noncarotid/nonlacunar.[6] The standardized criteria for stroke have remained consistent in ARIC over the entire study follow-up period.[6]

Embolic Risk Factors

Embolic risk factors for AIS were considered as potential mediators of the hypothesized effect and were collected for this study, most proximal to, but preceding the AIS event, unless otherwise specified. To determine the potential mediators on the association between age at stroke-onset and AIS subtype, measures of cardiac disease states with high embolic risk (AF,

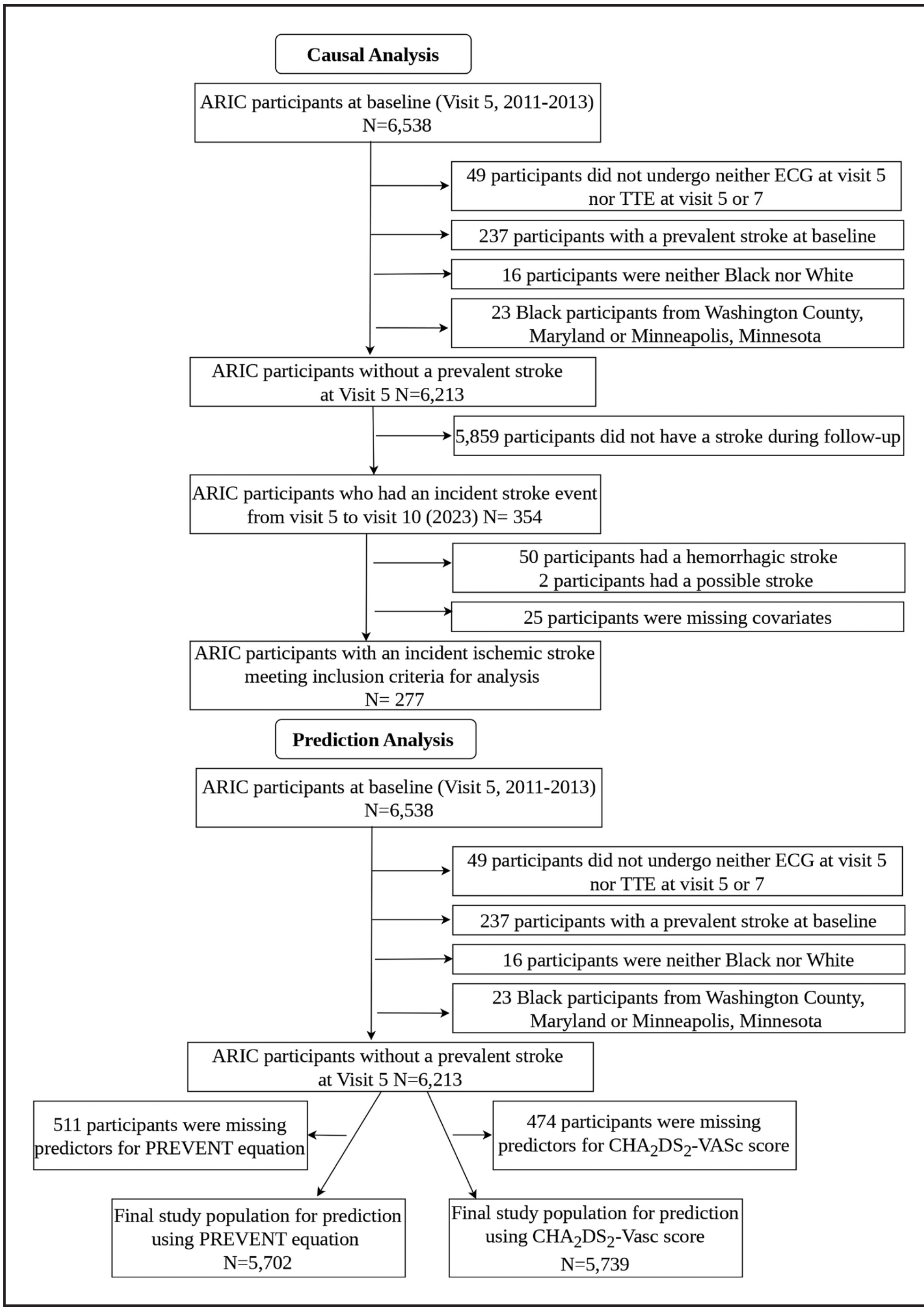

Figure 1. Study flowchart of participants included for analysis.
ARIC indicates Atherosclerosis Risk in Communities; PREVENT, Predicting Risk of Cardiovascular Disease Events; and TTE, transthoracic echocardiogram.

heart failure, coronary heart disease, myocardial infarction), and measures of left atrial structure and function (left atrial volume index [LAVI], left ventricular hypertrophy, P-wave amplitude in lead v1, P-wave duration in lead v1, P-wave axis) were chosen based on a priori hypotheses.

Cardiac events in ARIC have been described in detail in previous manuscripts.[12–16] In brief, AF is determined from study visit ECG (v1–5), hospital discharge records, and death certificates.[12,13] Heart failure is adjudicated in ARIC and is first determined from annual and semiannual telephone interviews, hospitalization records, and death certificates.[14] After detailed abstractions of the records, each hospitalization is then reviewed by 2 centrally trained and certified physicians, and the type of heart failure is then further classified as has been detailed previously.[14] Coronary heart disease is based on indications of fatal coronary heart disease, coronary revascularizations, or myocardial infarction from annual and semiannual telephone interviews, hospitalization records, and death certificates.[15] Myocardial infarction is also adjudicated in ARIC and is first determined from hospital discharge records, ECG evidence, chest pain symptoms, and biomarkers.[16] Trained physicians of the ARIC Mortality and Morbidity Classification Committee then review these after a computer-generated algorithm to determine myocardial infarction events based on standardized protocols.[16]

LAVi (mL/m^2) is obtained from transthoracic echocardiography, which in ARIC was performed at visits 5 and 7, and the results from the closest reading to, but preceding, the AIS event are used. LAVi were derived from left atrium volume measured using apical 4- and 2-chamber view at end-systolic frame before mitral valve opening and indexed to body surface area.[17]

Left ventricular hypertrophy and 3 measures of cardiac function, including P-wave amplitude in lead v1 (mV), P-wave duration in lead v1 (s), and P-wave axis (degrees), were obtained from ECG only at visit 5.

Covariates

Potential confounders were determined based on scientific knowledge, and the same confounders were used for causal and mediation analyses. Self-reported sex and combined race and center were collected in ARIC at visit 1. Other study covariates were collected most proximal to, but preceding the AIS event including hypertension (systolic blood pressure ≥140 mm Hg, diastolic blood pressure ≥90 mm Hg, or pharmacological treatment for hypertension), diabetes (hemoglobin A1C value ≥6.5% or pharmacological treatment for diabetes or self-report diagnosis of diabetes by a physician), current smoking, alcohol use, body mass index (kg/m^2), and LDL (low-density lipoprotein) cholesterol (mg/dL).

For the prediction analysis, variables that composed the original PREVENT equation,[4] and the CHA_2DS_2-VASc score,[18] with the exclusion of prevalent stroke, as this analysis is of incident stroke, were collected in ARIC at visit 5. Detailed information regarding included covariates is provided in Table S1.

Statistical Analysis

Characteristics between participants who had an AIS ≥80 years and <80 years during follow-up were compared using *t* tests or χ^2 tests as appropriate.

Causal Analysis

Age at AIS was ascertained from the summation of visit 5 age and follow-up time in days from visit 5 to stroke diagnosis. Given age at AIS was tied to AIS diagnosis, multivariable logistic regression models were used rather than time-to-event analysis to determine the association between age at AIS (≥80 versus <80 years) and AIS subtype (EIS versus TIS), with adjustment for potential confounders (see Covariates section). Secondary analyses further determined the association between age at AIS and EIS versus each TIS subtype (lacunar, carotid, or noncarotid/nonlacunar) in separate models. An interaction term (age at AIS×sex or age at AIS×race) was included when appropriate to explore differences by sex or race in the association between age at AIS and AIS subtype.

Continuous embolic risk factors were dichotomized based on meaningful clinical normative values: LAVi (reference: ≤34 mL/m^2), P-wave amplitude in lead v1 (reference: 0.05–0.25 mV), P-wave duration in lead v1 (reference: 0.08–0.10 s), and P-wave axis (reference: 0–75 degrees).[19,20] Logistic regression with bootstrapping (1000-iterations) under the counterfactual causal mediation framework was used to examine whether the effect of age at stroke-onset on AIS subtype was through individual embolic risk factors, with adjustment for potential confounders. The framework is under the standard identification assumptions, including consistency, no mediator-outcome confounder affected by exposure, and no unmeasured confounding of the exposure-outcome, exposure-mediator, and mediator-outcome relationships.[21] Each embolic risk factor was considered separately in distinct models, as the presence of one cardiac condition may increase the risk of having others.

Stroke Prediction

Receiver operating characteristic curves and C statistics were used to define the predictive performance for AIS among the oldest-old using the PREVENT equation. We used the PREVENT–atherosclerotic cardiovascular disease model to calculate AIS risk, as this is the model designed to assess atherosclerotic cardiovascular disease risk, which includes stroke. We then introduced our hypothesized embolic risk factors as covariates into the PREVENT–atherosclerotic cardiovascular disease model to determine if there was an improvement in the predictive performance. In addition to the PREVENT equation, we also assessed stroke risk using the CHA_2DS_2-VASc score to enhance the clinical meaningfulness of our analysis. The CHA_2DS_2-VASc score for each participant was calculated and categorized as 0, 1, 2, and ≥3. The original stroke risk was derived using a logistic regression model by including CHA_2DS_2-VASc categories. Then, a multivariable logistic regression model added the embolic risk factors as covariates to derive new stroke risk scores to compare the prediction before and after including these embolic risk factors. Finally, overall AIS prediction postvisit 5 among all ARIC participants was performed to give the prediction in the oldest-old context. We then further limited our prediction to EIS alone as the CHA_2DS_2-VASc scoring system focused on AF related risk, and AF is one of the leading causes of EIS.[22,23]

Multiple imputation with chained equation (m=10, maxit=15) was conducted for AIS prediction to impute missing data under a missing-at-random assumption, to ensure the validity of the results from the complete-case analysis.

Statistical analyses were performed using Stata v.18.0.[24] *P*≤0.05 was considered significant for all tests, except *P* interaction significance was set at *P*≤0.1.

RESULTS

There were 6213 stroke-free ARIC participants at visit 5; 277 of these participants developed an AIS (99 EIS, 178 TIS), with a median (Q1–Q3) time to develop AIS of 5.1 (2.6–7.1) years (Figure 1). Among those with an AIS, the median (Q1–Q3) age of participants at visit 5 was 76 (72–80), and their median age at AIS was 81 (77–86) years. Participants with AIS ≥80 years had lower body mass index (28.2 versus 30.0; *P*=0.02) and were less likely to be Black individuals (17.3% versus 34.9%; *P*<0.001; Table 1). Those who had an AIS ≥80 years were more likely to have AF (17.8% versus 6.4%; *P*=0.006), an abnormal LAVi (33.3% versus 20.2%;

Table 1. Study Participants Characteristics for Patients With Stroke, by Age at Incident Ischemic Stroke*

	Participants with incident stroke <80 y	Participants with incident stroke ≥80 y	
	(n=109)	(n=168)	*P* value
Participant's age at visit 5	72 (70–74)	80 (77–82.5)	<0.001
Female	55 (50.5%)	104 (61.9%)	0.06
Black race	38 (34.9%)	29 (17.3%)	<0.001
Body mass index, kg/m^2	30.0 (6.5)	28.2 (6.0)	0.02
Diabetes	43 (39.4%)	59 (35.1%)	0.47
Hypertension	90 (82.6%)	140 (83.3%)	0.87
LDL cholesterol level, mg/dL	113.1 (48.2)	103.3 (34.9)	0.051
Alcohol use	54 (49.5%)	74 (44.0%)	0.37
Current smoking	10 (9.2%)	7 (4.2%)	0.09
Atrial fibrillation	7 (6.4%)	30 (17.8%)	0.006
Coronary heart disease	22 (20.6%)	48 (28.6%)	0.12
Heart failure	27 (24.8%)	42 (25.0%)	0.97
Myocardial infarction	24 (22.0%)	30 (17.9%)	0.39
Left atrium volume index†	22 (20.2%)	56 (33.3%)	0.02
Left ventricular hypertrophy	15 (13.8%)	38 (22.6%)	0.07
P-wave duration in lead v1‡	105 (96.3%)	162 (96.4%)	0.97
P-wave amplitude in lead v1§	81 (74.3%)	132 (78.6%)	0.41
P-wave axis‖	14 (12.8%)	51 (30.4%)	<0.001

SI convention factors: To convert LDL cholesterol level to millimoles per liter, multiply by 0.0259. LDL indicates low-density lipoprotein.

*With the exception of sex and Black race, other characteristics are most proximal to, but preceding the incident ischemic stroke event. Univariate differences by age at incident ischemic stroke were calculated using *t* test for continuous variables, and χ^2 test for categorical variables. Continuous variables are displayed as mean (SD) except age at visit 5 is displayed as median (Q1–Q3); Categorical variables are displayed as n (%).

†Left atrial volume index reference value set to ≤34 mL/m^2.

‡P-wave duration, lead v1 reference range 0.08–0.10 s.

§P-wave amplitude, lead v1 reference range 0.05–0.25 mV.

‖P-wave axis reference range 0–75 degrees.

P=0.02), or an abnormal P-wave axis (30.4% versus 12.8%; *P*<0.001).

Association Between Age at AIS and Stroke Subtype

When examining the differences in AIS subtype by age at AIS, participants who had an AIS ≥80 years had a higher odds of EIS (versus TIS; odds ratio, 1.90 [95% CI, 1.09–3.31]), as compared with those who had a stroke <80 years (Table 2). When considering TIS subtype, there was a higher odds of EIS versus lacunar stroke among those ≥80 years (odds ratio, 2.18 [95% CI, 1.04–4.59]), but no statistically significant difference when comparing EIS to noncarotid/nonlacunar (odds ratio, 1.63 [95% CI, 0.87–3.05]). We also found a higher odds of EIS versus carotid artery stenosis–related stroke among those ≥80 years (odds ratio, 9.22 [95% CI, 1.40–60.51]). However, results were imprecise due to the small number of participants with carotid artery stenosis–related stroke (n=8).

Table 2. The Association Between Age at Incident Ischemic Stroke (≥80 Versus <80 Years) and Incident Ischemic Stroke Subtype (EIS Versus TIS)

	Univariate	Adjusted*
	OR (95% CI)	OR (95% CI)
EIS (vs any TIS; n=277)	2.12 (1.25–3.59)	1.90 (1.09–3.31)
EIS (vs lacunar; n=165)	2.11 (1.10–4.06)	2.18 (1.04–4.59)
EIS (vs carotid; n=107)	7.61 (1.45–39.97)	9.22 (1.40–60.51)
EIS (vs nonlacunar/carotid; n=203)	1.93 (1.08–3.47)	1.63 (0.87–3.05)

EIS indicates embolic ischemic stroke; LDL, low-density lipoprotein; OR, odds ratio; and TIS, thrombotic ischemic stroke.

*Model adjusted for sex (female vs male), Black race, and risk factors most proximal to, but preceding the incident ischemic stroke event, including body mass index, alcohol use, current smoking, LDL cholesterol level, hypertension, and diabetes.

No differences were found between age at AIS and subtype by sex (*P* interaction=0.29; Table S2) or race (*P* interaction=0.97; Table S3).

Mediation of the Effect of Age at AIS on AIS Subtype Through Embolic Risk Factors

To determine if the association between age at AIS and EIS was mediated by embolic risk factors, we tested the (1) indirect effects of age at AIS on stroke subtype through each embolic risk factor, (2) direct effects of age at AIS on stroke subtype through mechanisms other than the individual embolic risk factor, (3) total effect (indirect plus direct) of age on stroke subtype (effect estimates shown in Table S4). The proportion of the total effect of age at AIS on EIS (defined as indirect effect/total effect) was 44% mediated by AF (0.0602/0.1365; *P*=0.03), 45% mediated by abnormal LAVi (0.0602/0.1342; *P*=0.048), and 43% mediated by abnormal P-wave axis (0.0595/0.1369; *P*=0.04; Table 3).

Prediction of AIS

To determine if the inclusion of the embolic risk factors improves AIS prediction, we compared the C statistics of the original model (calculated using the PREVENT equation, or the CHA_2DS_2-VASc score) to the extended model (inclusion of embolic risk factors; Tables S5 and S6). Receiver operating characteristic curves and associated C statistics were shown in Figure 2 for AIS prediction among the oldest-old, Figure S1 for AIS prediction postvisit 5 among all ARIC participants, and Figure S2 for EIS prediction postvisit 5 among all ARIC participants and among the oldest-old.

Overall, 5702 participants had complete covariate information for the PREVENT equation, and 5739 participants had complete covariate information for the CHA_2DS_2-VASc score (Figure 1).

The PREVENT equation C statistic for AIS risk among the oldest-old was 0.49 (95% CI, 0.45–0.53; Table S5).

Table 3. Proportion of the Effect Mediated by Each Embolic Risk Factor in the Difference in Odds of Embolic Ischemic Stroke (Versus Thrombotic Ischemic Stroke) by Age at Incident Ischemic Stroke (≥80 Versus <80 Years; N=277)*

Mediator	Proportion of the effect mediated by the embolic risk factor Proportion (95% CI)†	*P* values
Atrial fibrillation	0.44 (0.04 to 0.84)	0.03
Coronary heart disease	0.001 (−0.08 to 0.08)	0.98
Heart failure	0.06 (−0.09 to 0.21)	0.43
Myocardial infarction	0.004 (−0.06 to 0.06)	0.90
Left atrial volume index, mL/m²‡	0.45 (0.004 to 0.89)	0.048
Left ventricular hypertrophy	0.05 (−0.08 to 0.19)	0.42
P-wave amplitude, lead v1, mV§	−0.002 (−0.08 to 0.07)	0.97
P-wave duration, lead v1, s‖	−0.001 (−0.06 to 0.06)	0.98
P-wave axis, degrees¶	0.43 (0.01 to 0.86)	0.04

LDL indicates low-density lipoprotein.

*Mediators were most proximal to, but preceding the incident stroke event, P-wave duration, lead v1, P-wave amplitude, lead v1, P-wave axis, and left ventricular hypertrophy were obtained at visit 5; left atrial volume index was obtained at visit 5 and 7, and the closest reading of the index to the incident ischemic stroke event (but before the incident ischemic stroke) was used.

†Mediation model adjusted for sex (female vs male), Black race, and risk factors most proximal to, but preceding the incident ischemic stroke event, including body mass index, alcohol use, current smoking, LDL cholesterol level, hypertension, and diabetes. The proportion of the total effect of age at incident ischemic stroke on embolic ischemic stroke was defined as the indirect effect/total effect.

‡Left atrial volume index reference value set to ≤34 mL/m².

§P-wave amplitude, lead v1 reference range 0.05–0.25 mV.

‖P-wave duration, lead v1 reference range 0.08–0.10 s. Mediator model for p-wave duration, lead v1(s) did not adjust for current smoking due to perfect prediction.

¶P-wave axis reference range 0–75 degrees.

Addition of the embolic risk factors that we found to be significant mediators (AF, LAVi, and P-wave axis) improved the model performance (C statistic, 0.77 [95% CI, 0.74–0.80]; *P*<0.001), and inclusion of all hypothesized embolic risk factors slightly increased model performance (C statistic, 0.79 [95% CI, 0.76–0.82]; *P*<0.001). In addition, the C statistic for the original PREVENT equation when predicting EIS among the oldest-old was 0.50 (95% CI, 0.44–0.57), and the model performance greatly improved for the extended model with 3 embolic risk factors (C statistic, 0.84 [95% CI, 0.80–0.88]; *P*<0.001).

The CHA_2DS_2-VASc score C statistic for AIS risk among the oldest-old was 0.57 (95% CI, 0.55–0.59; Table S6). The C statistic of 0.63 (95% CI, 0.59–0.67) for the extended model with the embolic risk factors that we found to be significant mediators (AF, LAVi, P-wave axis) showed only subtle improvement (*P*<0.001). EIS prediction among the oldest-old was further examined with a C statistic of 0.57 (95% CI, 0.54–0.60) for the original CHA_2DS_2-VASc score, and better model performance was shown when using the extended model with 3 embolic risk factors (C statistic, 0.72 [95% CI, 0.66–0.78]; *P*<0.001).

Sensitivity Analysis for AIS Prediction

Participants' demographics and clinical characteristics by missing variable status for the PREVENT equation and CHA_2DS_2-VASc score are shown in Tables S7 and S8. When prediction variables were imputed, the results did not differ as compared with our complete-case analysis (Tables S9 and S10).

DISCUSSION

In this analysis of older adults living in the community aged 67 to 90 years, those with an incident AIS at age ≥80 were more likely to have EIS, compared with those who had a first AIS <80 years, with no difference by sex or race. We also found that AF, LAVi, and P-wave axis mediated the effect of age at AIS on AIS subtype, and the relative contribution for each was almost half of the effect. Notably, both the PREVENT equation and the CHA_2DS_2-VASc score had relatively poor performance in predicting stroke, especially among the oldest-old. Adding those embolic risk factors that mediated the effect improved the AIS prediction, yet adding all embolic risk factors improved AIS prediction only slightly.

Although the risk of EIS is known to increase with age,[25,26] the appropriateness of extrapolating this data to the oldest-old was questionable, as most studies included individuals with a mean age of 65 years.[27] We now provide a unique perspective of examining AIS at even older ages; we have now shown that the odds of having EIS (versus TIS) increased by 90% when participants experienced an AIS event at age ≥80.

Another critical perspective in considering the stroke etiology among the oldest-old is to examine associated risk factors. AF, heart failure, and myocardial infarction are established independent risk factors for EIS,[22,27] and prior research indicated that patients with EIS, as compared with TIS, were associated with more severe stroke and a higher risk of death,[28] suggesting an important role of cardiac embolism in stroke etiology. Given that advancing age is associated with an increased risk of cardiac conditions, and people with cardiac conditions have higher stroke incidence,[29,30] these conditions may represent key mediators of the effect of age on stroke subtype. Our findings showed that AF, abnormal LAVi, and abnormal P-wave axis—markers of cardiac structural and electrical abnormalities such as left atrial enlargement—mediated the association between age at stroke-onset and AIS subtype. The trend that adults experiencing an AIS at age ≥80 years are more likely to present with EIS is likely due to the accumulation of embolic risk factors that extend beyond traditional vascular risk profiles. We, therefore, suggest that routine screening for cardiac disease is imperative as it allows early interventions to prevent cardiac morbidities and stroke as people age.

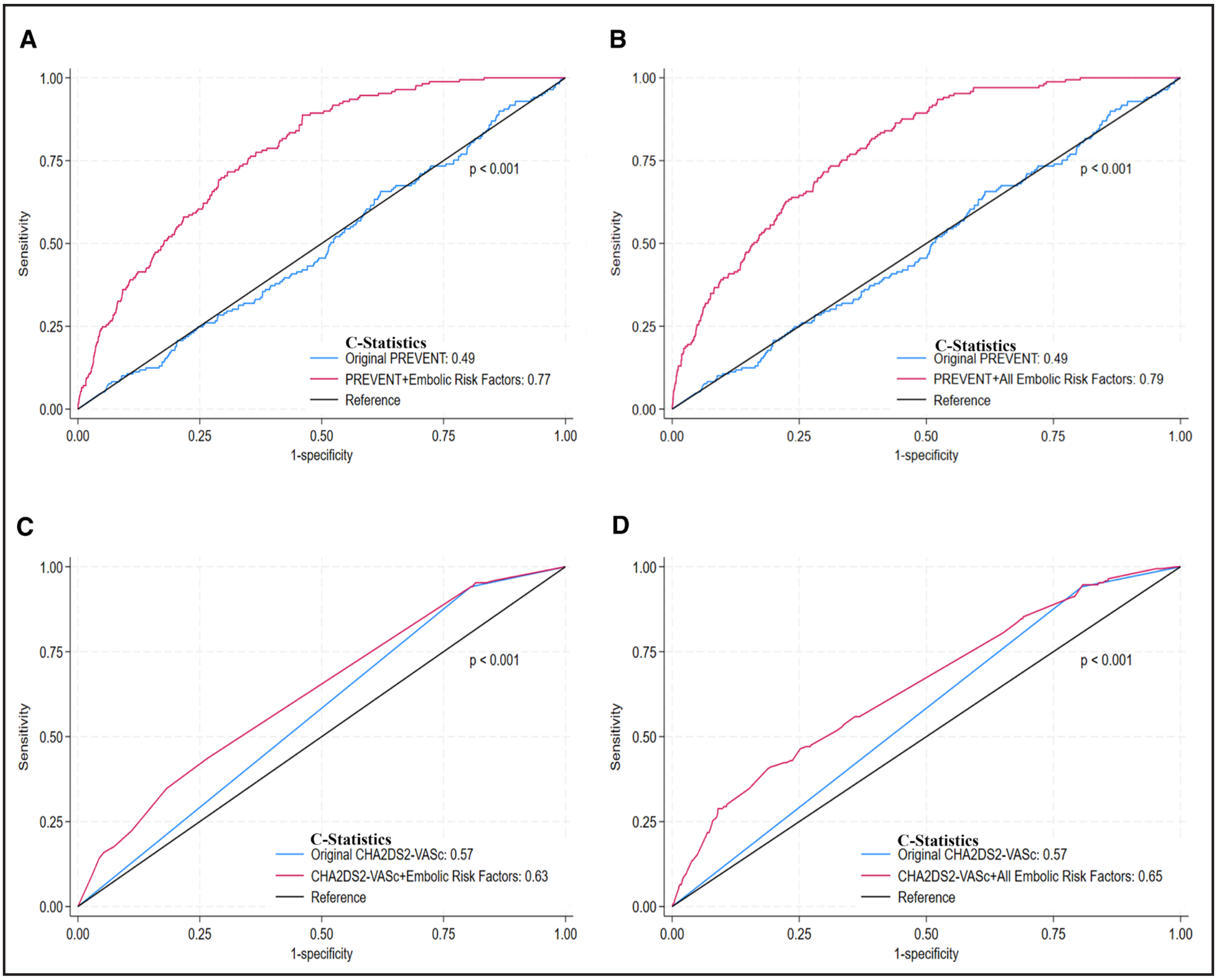

Figure 2. Incident ischemic stroke prediction among age ≥80 using the Predicting Risk of Cardiovascular Disease Events (PREVENT) equation (n=5702) and the CHA_2DS_2-VASc score (n=5739).

Receiver operating characteristic (ROC) Curves and associated C statistics were used to demonstrate incident ischemic stroke prediction among age ≥80 years using PREVENT equation or CHA_2DS_2-VASc score. $P \leq 0.05$ was considered significant. **A**, Comparing the predictive performance of stroke prediction among age ≥80 years using the original PREVENT–atherosclerotic cardiovascular disease (ASCVD) model vs the PREVENT-ASCVD in addition to embolic risk factors that were found to be significant mediators, including atrial fibrillation, left atrial volume index, P-wave axis. **B**, Comparing the predictive performance of stroke prediction among age ≥80 years using the original PREVENT-ASCVD model vs the PREVENT-ASCVD in addition with hypothesized embolic risk factors, including atrial fibrillation, left atrial volume index, coronary heart disease, myocardial infarction, heart failure, left ventricular hypertrophy, P-wave duration lead v1, P-wave amplitude lead v1, P-wave axis. **C**, Comparing the predictive performance of stroke prediction among age ≥80 years using CHA_2DS_2-VASc categories (derived using logistic regression) vs CHA_2DS_2-VASc categories in addition to embolic risk factors that were found to be significant mediators, including atrial fibrillation, left atrial volume index, P-wave axis. **D**, Comparing the predictive performance of stroke prediction among age ≥80 years using CHA_2DS_2-VASc categories (derived using logistic regression) vs CHA_2DS_2-VASc categories in addition with hypothesized embolic risk factors, including atrial fibrillation, left atrial volume index, coronary heart disease, left ventricular hypertrophy, P-wave duration lead v1, P-wave amplitude lead v1, P-wave axis. [a]Embolic risk factors that were found to be significant mediators, including atrial fibrillation, left atrial volume index, and P-wave axis.

The PREVENT equation or the CHA_2DS_2-VASc score may not be optimal stroke prediction tools, but they represent some of the best tools currently available to clinicians for stroke prediction. Neither the newly developed PREVENT equation nor the widely used CHA_2DS_2-VASc score focused on stroke-specific prediction, not to mention late-life stroke. The PREVENT equation considers stroke as a composite outcome within atherosclerotic cardiovascular disease, and the CHA_2DS_2-VASc score focuses on AF related stroke risk.[4,18] We observed relatively low C statistics when predicting AIS, or EIS alone, using both tools. Prediction performances were observed to be better when adding embolic risk factors, especially AF, LAVi, and P-wave axis into the model, with greater improvements when predicting late-life stroke risk. Although the CHA_2DS_2-VASc score is primarily focused on stroke prevention among patients with AF, better predictive performances were still observed among the oldest-old, irrespective of AF status. These emphasize the importance of embolic risk factors when developing stroke-specific prediction tools that could be used for effective stroke prevention. Future research could also consider incorporating treatment of cardiac conditions when assessing stroke risk, as this information may be important but was not available in our study.

The study has limitations. First, ARIC is a visit-based cohort study, with certain data collected only at in-person visits, but with ongoing continuous surveillance for stroke events. This means the actual status of risk factors included as confounders or mediators at the precise time of the AIS event may not be captured. However, we incorporated risk factor data before the AIS event, and given regular visits during this study period in ARIC, as proximal to the stroke event as possible, to ensure relevance. Second, since once conditions like hypertension are diagnosed, although they may be treated, the condition is unlikely to be completely reversed. Third, our study only included White and Black participants, given the low proportions of other races at the initial ARIC recruitment. Therefore, results may be different when including other races and ethnicities. Fourth, we recognize that some ARIC participants did not undergo ECG or transthoracic echocardiography, and therefore, our analysis of those who did may represent a different population or have some intrinsic differences from those who did not complete these tests. However, ≈94% of ARIC participants at visit 5 did complete ECG or transthoracic echocardiography, thereby reducing the likelihood that we selected a different population due to nonparticipation in these tests. Fifth, it is possible that there is a chance of type I error due to failure to adjust for multiple comparisons in our mediation analysis; however, we did make this decision intentionally, given the overarching desire to determine potential mediators on age at stroke-onset and AIS subtype. We view our work as an exploration of potential mediators and, therefore, did not adjust for multiple comparisons. Sixth, the presence of AF is considered in the ARIC adjudication process for EIS. However, EIS is classified based on a composite adjudication emphasizing embolic imaging features as well as the exclusion of competing mechanisms. AF, when present, is considered supportive but is neither necessary nor sufficient for EIS classification. If AF were completely deterministic of EIS, then it would not be appropriate to consider it a mediator in our analysis. However, 71% of our cases of EIS had no documented AF, and the variance inflation factor was 1.14, suggesting that AF and EIS were not colinear. Another limitation is that this study focused solely on incident stroke in late life and, therefore, does not address recurrent stroke. Silent infarcts are also common at older ages, and this study does not include silent infarcts. Lastly, we recognized that the number of participants who are aged >80 years is small; however, the number of these participants in ARIC is quite robust compared with other clinical or population cohort studies. This is likely one reason why the oldest-old remain underrepresented in current health research fields. Thus, we think that this adds to the novelty of our findings.

Our exploration of stroke etiology among the oldest-old revealed that individuals experiencing a first-ever stroke at advanced ages were more likely to have an EIS, and this association was mediated through AF, abnormal LAVi, or abnormal P-wave axis. These findings highlight the importance of not only addressing the increased risk of stroke with aging but also prioritizing early detection and prevention of underlying cardiac conditions. The PREVENT equation and CHA_2DS_2-VASc score performed poorly in predicting stroke risk; however, incorporating embolic risk factors improved predictive performance, although these tools were neither developed specifically for stroke nor applied to oldest-old. Future research should explore additional cardiac conditions that may contribute to EIS in aging populations and focus on developing stroke-specific risk prediction models to better guide prevention strategies.

ARTICLE INFORMATION

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Affiliations

Department of Neurology (J.W., M.C.J.) and Department of Cardiology (C.E.N.), The Johns Hopkins University School of Medicine, Baltimore, MD. Cognition and Neuroepidemiology Section, National Institute of Neurological Disorders and Stroke (NINDS), Bethesda, MD (M.E., R.F.G.). Department of Epidemiology, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD (Z.J.). Division of Epidemiology and Community Health, School of Public Health, University of Minnesota, Minneapolis (K.L.). Department of Population Health, New York University Grossman School of Medicine (J.C.).

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Supplemental Material

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